25900367 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECC202 - NETWORKING FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. State the unit of bandwidth. [K] / [CO1]
2. Show how five computers can be connected in a ring topology. [K] / [CO1]
3. Describe bit interval and bit rate. [U] / [CO1]
4. Differentiate between attenuation and distortion. [An] / [CO1]
5. What is meant by virtual-circuit network? [K] / [CO2]
6. Differentiate between Stop-and-Wait ARQ and Go-Back-N ARQ with respect to any one key feature. [U] / [CO3]
7. Recall which protocol is considered the basis for Ethernet. [K] / [CO3]
8. Differentiate between ACK and NACK with respect to error control. [U] / [CO3]
9. State how many layers are in OSI. [K] / [CO1]
10. Draw a simple diagram showing how a bridge connects two LAN segments. [K] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

11.	Compare FTP and HTTP for file transfer in simple terms.	[An] / [CO1]
12.	Explain three types of guided media.	[U] / [CO2]
13.	Name and briefly describe the function of at least three common application layer protocols	[K] / [CO4]
14.	Indicate the IEEE standard used for Gigabit Ethernet and mention the type of cabling it supports.	[K] / [CO3]
15.	Differentiate between stop - and- wait protocol and sliding window protocol	[U] / [CO4]
16.	Explain how a firewall can prevent unauthorized access.	[U] / [CO4]
17.	Explain any two techniques used for congestion control.	[U] / [CO4]
18.	Distinguish between static IP addressing and dynamic IP addressing in both IPv4 and IPv6.	[U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

19.	Analyse the TCP/IP model by explaining the functions of each layer and associating common protocols with them.	[An] / [CO1]
20.	Given a wireless system with limited bandwidth, how would you apply spread spectrum techniques to ensure that multiple users can access the network efficiently?	[A] / [CO2]
21.	Demonstrate with the help of a block diagram how a call is established and maintained in a cellular network, including the role of base stations and mobile switching centers.	[U] / [CO3]

[10x2 = 20]